



Lecture 07 - Programs & APPS



06016402 : Information Technology Fundamentals

Asst.Prof.Dr. Pornsuree Jamsri

1. Programs and APPs

A **program**, or **software**, consists of a series of related instructions, organized for a common purpose, that tells the computer what tasks to perform and how to perform them.

โปรแกรม หรือ ซอฟต์แวร์ ประกอบไปด้วยชุดคำสั่งที่เกี่ยวข้องกัน ซึ่งพัฒนาขึ้นเพื่อวัตถุประสงค์ร่วมกัน เพื่อบอกคอมพิวเตอร์ว่า มีสิ่งที่จะต้องทำอะไรบ้าง และบอกถึงวิธีการ ขั้นตอนต่างๆ ที่คอมพิวเตอร์จะต้องทำอย่างไร

1. Programs and APPs

- An **application**, or **app**, sometimes called application software, consists of programs designed to make users more productive and/or assist them with personal tasks
- Application หรือ app บางครั้งเรียกว่า Application software ประกอบไปด้วย โปรแกรมที่ถูกออกแบบมาเพื่อให้ **user** ใช้งานได้อย่างมีประสิทธิภาพ และ/หรือ ออกแบบมาเพื่อช่วย **user** ในงานส่วนตัวทั่วไปต่างๆ

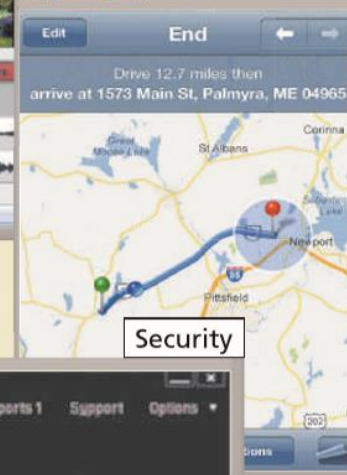
Productivity



Graphics and Media



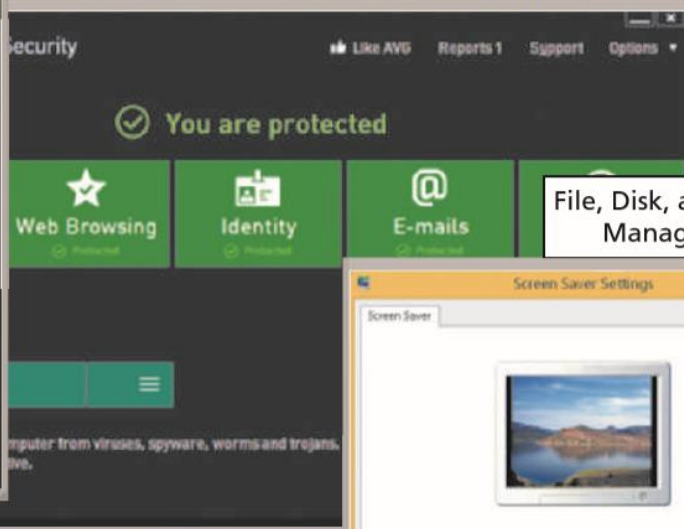
Personal Interest



Communications



Programs
and Apps



Security

File, Disk, and System
Management

Operating system

- An **operating system** is a set of programs that coordinates all the activities among computer or mobile device hardware
- The operating system, therefore, **serves as the interface between the user, the applications** and other programs, and the computer's or mobile device's **hardware**.

ดังนั้น ระบบปฏิบัติการ ทำหน้าที่ให้บริการ การเชื่อมต่อ ระหว่าง **user** กับ **software** และ **hardware**.

An Example of How an Operating System Interacts with a User, an Application, and Hardware

Step 1

An operating system user interface is displayed on the screen when a user turns on a computer.



Step 2

The user instructs the operating system to run an application, such as a word processing application, which appears on the user's screen. The user then instructs the application to print a document.

Print button

document to be printed



Step 4

The operating system sends the print instruction to the printer (hardware), which prints the document.



Step 3

The application sends the print instruction to the computer's operating system (software), which is running on the computer.



2. Obtaining software

- **Retail software** is mass-produced, copyrighted software that meets the needs of a wide variety of users, not just a single user or company.

ซอฟต์แวร์ที่วางจำหน่ายทั่วไป

- **Custom software** performs functions specific to a business or industry. Custom software usually costs more than retail software.

ซอฟต์แวร์แบบว่าจ้าง หรือ ซอฟต์แวร์แบบจ้างทำ จะทำงานตามฟังก์ชันที่ถูกกำหนดโดยผู้ว่าจ้าง จากภาคธุรกิจ อุตสาหกรรม.

2. Obtaining software

- **A web app** is an application stored on a web server that you access through a browser.
- **A mobile app** is an application you download from a mobile device's app store, sometimes called a marketplace, or other location on the Internet to a smartphone or other mobile device.
- **A mobile web app** is a web app that is optimized for display in a browser on a mobile device, regardless of screen size or orientation. Many app developers opt for web delivery because they do not have to create a different version for each mobile device's app store. Many web apps use a **responsive design**, which means the app displays properly on any computer or device

2. Obtaining software

- **Shareware** is copyrighted software that is distributed at **no cost for a trial period**. To use a shareware program beyond that period, you send payment to the software developer or you might be billed automatically unless you cancel within a specified period of time.
- **Freeware** is copyrighted software provided at no cost by an individual or a company that retains all rights to the software.
- **Open source software** is software provided for use, modification, and redistribution. This software has no restrictions from the copyright holder regarding modification of the software's internal instructions and its redistribution.
- **Public-domain software** has been donated for public use and has no copyright restrictions.

3. Categories of Programs and Apps

Table 4-1 Programs and Apps by Category

Category	Types of Programs and Apps	
Productivity (Business and Personal)	• Word Processing • Presentation • Spreadsheet • Database • Note Taking • Calendar and Contact Management • Project Management	• Accounting • Personal Finance • Legal • Tax Preparation • Document Management • Support Services • Enterprise Computing
Graphics and Media	• Computer-Aided Design (CAD) • Desktop Publishing • Paint/Image Editing • Photo Editing and Photo Management	• Video and Audio Editing • Multimedia and Website Authoring • Media Player
Personal Interest	• Lifestyle • Medical • Entertainment • Convenience • Education	

3. Categories of Programs and Apps

Communications

- Blog
- Browser
- Chat Room
- Online Discussion
- Email
- File Transfer
- Internet Phone
- Internet Messaging
- Mobile Messaging
- Videoconference
- Web Feeds

Security

- Personal Firewall
- Antivirus
- Malware Removers
- Internet Filters

File, Disk, and System Management

- File Manager
- Search
- Image Viewer
- Uninstaller
- Disk Cleanup
- Disk Defragmenter
- Screen Saver
- File Compression
- PC Maintenance
- Backup and Restore

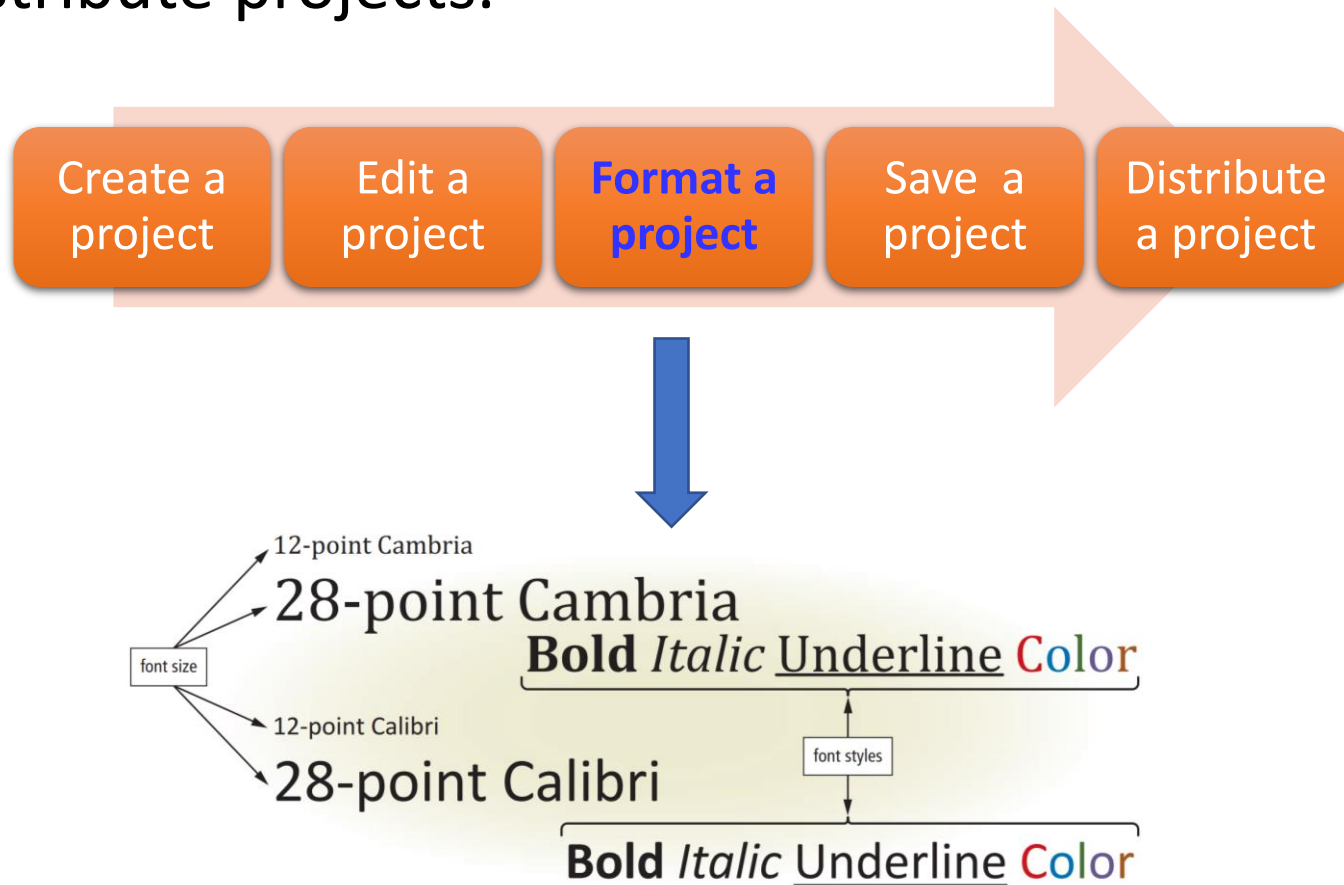
4. Productivity Applications

- **Productivity applications** can assist you in becoming more effective and efficient while performing daily activities at work, school, and home.
- Productivity applications include

Word processing	Presentation	Spreadsheet	Database	Note taking
Calendar and contact management	Project management	Accounting	Personal finance	Legal
	Tax preparation	Document management	Enterprise computing	

4. Productivity Applications

- **Developing Projects** With productivity applications, users often create, edit, format, save, and distribute projects.



4.1 Word Processing

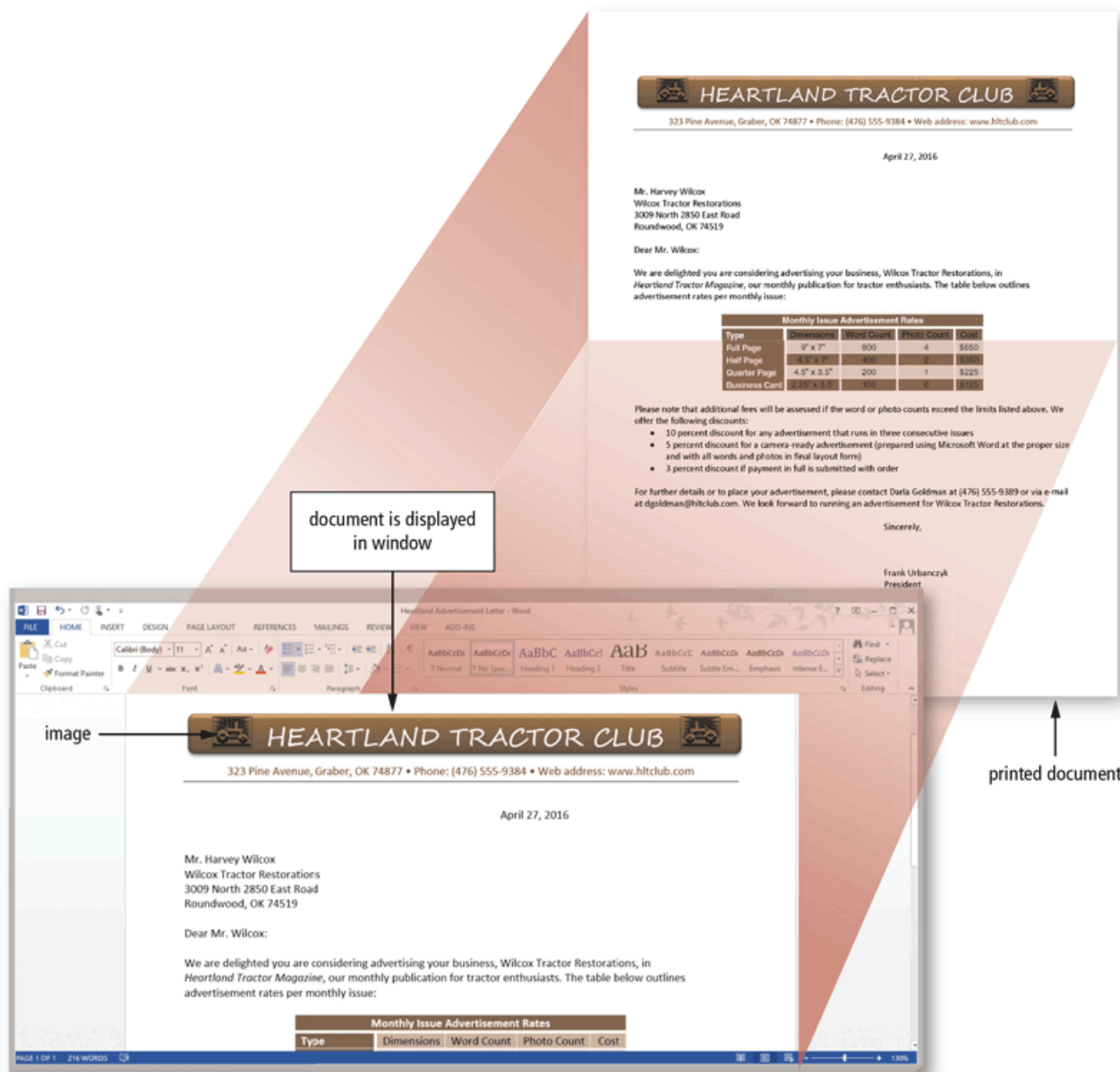


- **Word processing software**, sometimes called a **word processor**, is an application that allows users to create and manipulate documents containing mostly text and sometimes graphics.
- A major advantage of using word processing software is that it enables users to change their written words easily.

Word processing software also has many features to make documents look professional and visually appealing.

For example, you can change the font, size, and color of characters; apply special effects ; and organize text in newspaper-style columns.

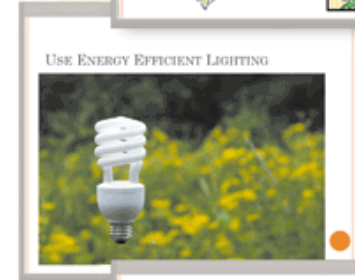
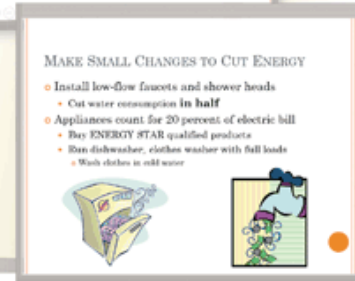
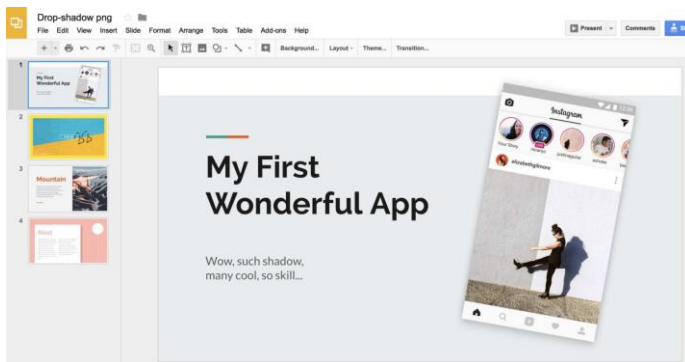
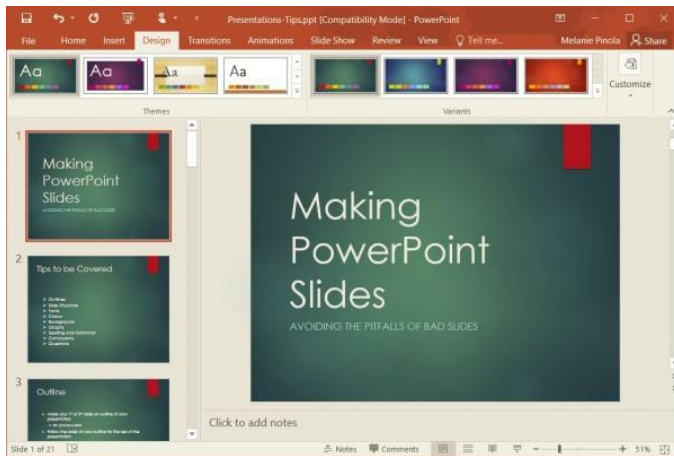
Word Processing



4.2 Presentation



- **Presentation software** is an application that allows users to create visual aids for presentations to communicate ideas, messages, and other information to a group.
- The presentations can be viewed as slides, sometimes called a **slide show**, that are displayed on a large monitor or on a projection screen from a computer or mobile device



4.3 Spreadsheet



Google Sheets

- **Spreadsheet software** is an application that allows users to organize data in columns and rows and perform calculations on the data. These columns and rows collectively are called a **worksheet**.
- **A cell is the intersection of a column and row**. The spreadsheet software identifies cells by the column and row in which they are located. For example, the intersection of column B and row 4 is referred to as cell B4.

formula assigned to active cell
cell contains a function

columns J, K, L

rows 3, 4,
5, and 6

=SUM(B9:B16)

cell B4

cell B17 is active

Bob Gremlin Budget

File Edit View Insert Format Data Tools Help All changes saved in Drive

Comments

Share

Personal Budget Worksheet

Monthly Estimates

	Income	January	February	March	April	May	June	July	August	September	October	November	December	Total
3	Wages	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$12,003.48
4	Dividends	4,000.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4,000.75	0.00	0.00	0.00	8,001.50
5	Total	\$5,001.04	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$1,000.29	\$5,001.04	\$1,000.29	\$1,000.29	\$1,000.29	\$20,004.98
6														
7														
8	Expenses	January	February	March	April	May	June	July	August	September	October	November	December	Total
9	Rent	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$400.89	\$4,810.68
10	Food	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	3,600.00
11	Tuition	1,500.00	0.00	0.00	0.00	0.00	0.00	0.00	1,500.00	0.00	0.00	0.00	0.00	3,000.00
12	Books	500.00	0.00	0.00	0.00	0.00	0.00	0.00	500.00	0.00	0.00	0.00	0.00	1,000.00
13	Entertainment	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	1,200.00
14	Car Payment	154.79	154.79	154.79	154.79	154.79	154.79	154.79	154.79	154.79	154.79	154.79	154.79	1,857.48
15	Gas	100.00	100.00	100.00	100.00	150.00	150.00	150.00	100.00	100.00	100.00	100.00	100.00	1,350.00
16	Miscellaneous	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	1,200.00
17	Total	\$3,155.68	\$1,155.68	\$1,155.68	\$1,155.68	\$1,205.68	\$1,205.68	\$1,205.68	\$3,155.68	\$1,155.68	\$1,155.68	\$1,155.68	\$1,155.68	\$18,018.16
18														
19	Net	\$1,845.36	(\$155.39)	(\$155.39)	(\$155.39)	(\$205.39)	(\$205.39)	(\$205.39)	(\$2,155.39)	\$3,845.36	(\$155.39)	(\$155.39)	(\$155.39)	\$1,986.82

Monthly Expenses Monthly Budget Planner

=SUM(B9:B16)

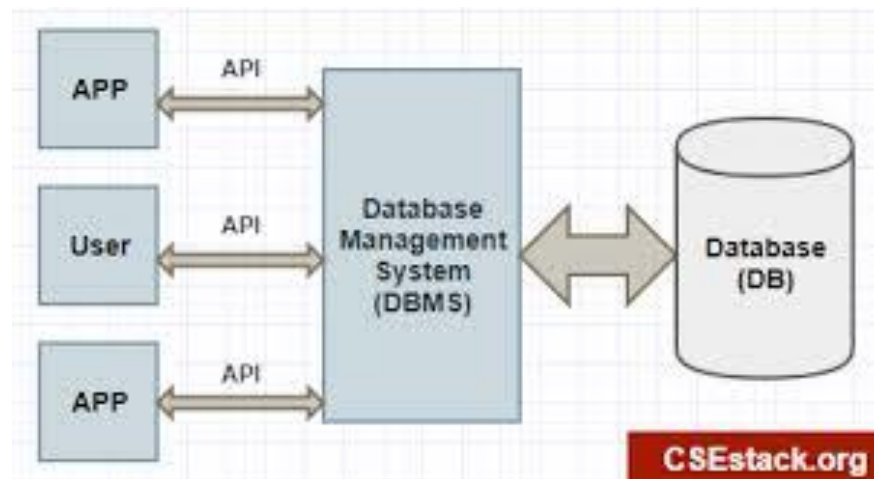
4.3 Spreadsheet

- A formula performs calculations on the data in the worksheet and displays the resulting value in a cell, usually the cell containing the formula. When creating a worksheet, you can enter your own formulas.
- A **function** is a predefined formula that performs common calculations, For example, the function =SUM(B9:B16).
- One of the more powerful features of spreadsheet software is its capability to recalculate the rest of the worksheet when data in a cell changes.
- Another standard feature of spreadsheet software is charting, which **depicts the data in graphical form**, such as bar charts or pie charts.

4.4 Database



- A **database** is a collection of data organized in a manner that allows access, retrieval, and use of that data.
- **Database software**, often called a **database management system (DBMS)**, is an application that allows users to create, access, and manage a database. Using database software, you can add, change, and delete data in a database; sort and retrieve data from the database; and create forms and reports using the data in the database.



4.4 Database

- With most personal computer database programs, a database consists of a **collection of tables**, organized in rows and columns. **Each row, called a record**, contains data about a given item in the database, which is often a person, product, object, or event. **Each column, called a field**, contains a specific category of data within a record.

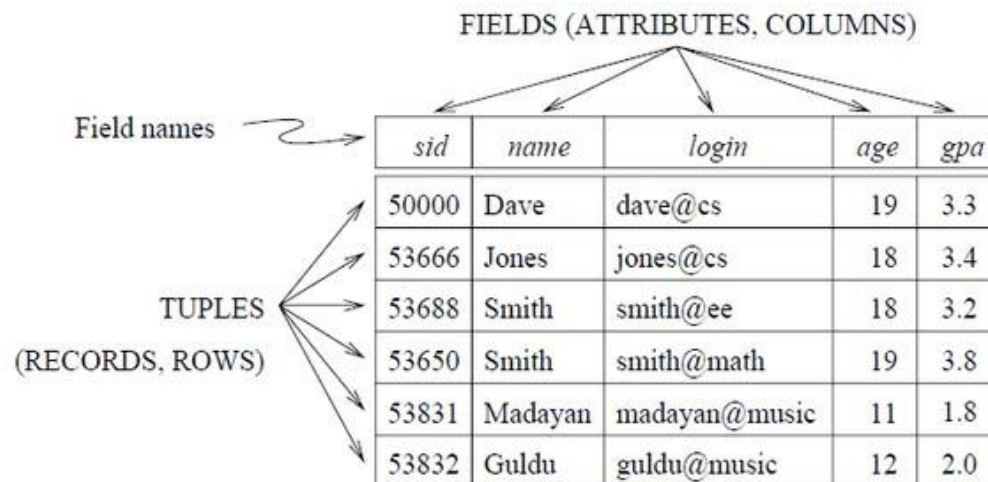


Figure 3.1 An Instance *S1* of the Students Relation

Customer table

fields

records

Book Rep table

common BR # field relates the Customer table to the Book Rep table

Customer

CU #	Customer Name	Street	City	State	Postal Code	Amount Paid	Current Due	Returns	BR #	Click to Add
ASU37	Applewood State University	300 University Ave.	Greer	PA	19158	\$41,530.98	\$38,812.66	\$12,008.02	42	
BCJ21	Brodikin Junior College	4806 Park Dr.	Greer	PA	19158	\$0.00	\$6,081.98	\$0.00	42	
CSD25	Cowpens ISD	829 Wooster Ave.	Gossett	PA	19157	\$12,750.00	\$13,275.00	\$2,000.00	53	
CSU10	Camellia State University	725 Camellia St.	Pleasantburg NJ	07025		\$63,246.88	\$69,847.76	\$14,967.98	53	
DCC34	Dartt Community College	3827 Burgess Dr.	Chambers	NJ	07037	\$21,345.50	\$23,467.75	\$4,496.80	65	
FSC12	First State College	7300 Cedar Rd.	Quaker	DE	19719	\$34,557.25	\$23,875.98	\$7,234.99	65	
FSU23	Farnham State University	1445 Hubert St.	Gaston	DE	19723	\$18,268.00	\$22,024.50	\$3,496.23	42	
HSD05	Hammonds County ISD	446 Lincoln Blvd.	Chesnee	NJ	07053	\$18,434.75	\$0.00	\$0.00	53	
KCC78	Key Community College System	7523 Penn Ave.	Adelphia	PA	19159	\$21,288.65	\$11,367.49	\$2,256.98	65	
MCC45	Mauldin Community College	72 Montrose St.	Chesnee	NJ	07053	\$9,500.00	\$5,000.00	\$0.00	53	
MEC56	Mehitable College	202 College St.	Gaston	DE	19723	\$9,111.19	\$7,310.76	\$574.25	42	
PLI22	PrattLast Institute	236 Ashton Ave.	Pleasantburg NJ	07025		\$17,229.45	\$11,769.75	\$3,450.50	53	
CC77	Stallone Community College	1200 Franklin Blvd.	Adelphia	PA	19156	\$7,525.98	\$2,515.78	\$0.00	42	
EC19	Seaborn College	345 Mather Rd.	Quaker	DE	19719	\$9,225.34	\$10,496.89	\$2,017.67	65	
SC02	Tri-State Consortium	3400 Metro Pkwy.	Adelphia	PA	19156	\$34,578.90	\$9,432.56	\$943.34	65	

Book Rep

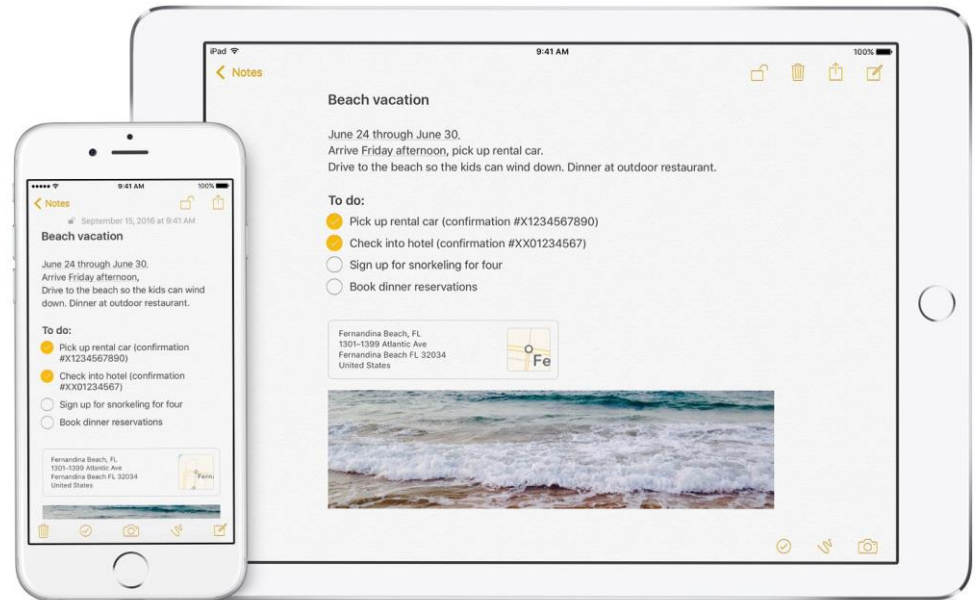
BR #	Last Name	First Name	Street	City	State	Postal Code	Start Date	Salary	Bonus Rate	Click to Add
42	Perez	Melina	261 Porter Dr.	Adelphia	PA	19156	5/14/2012	\$31,500.00	0.20	
48	Statnik	Michael	3135 Simpson Dr	Pleasantburg	NJ	07025	1/15/2013	\$29,000.00	0.20	
53	Chin	Robert	265 Maxwell St.	Gossett	PA	19157	6/1/2013	\$26,250.00	0.19	
65	Rogers	Tracy	1827 Maple Ave.	Adelphia	PA	19159	7/1/2014	\$7,750.00	0.18	
*								\$0.00		

Customer Number (three uppercase letters followed by 2-digit number)

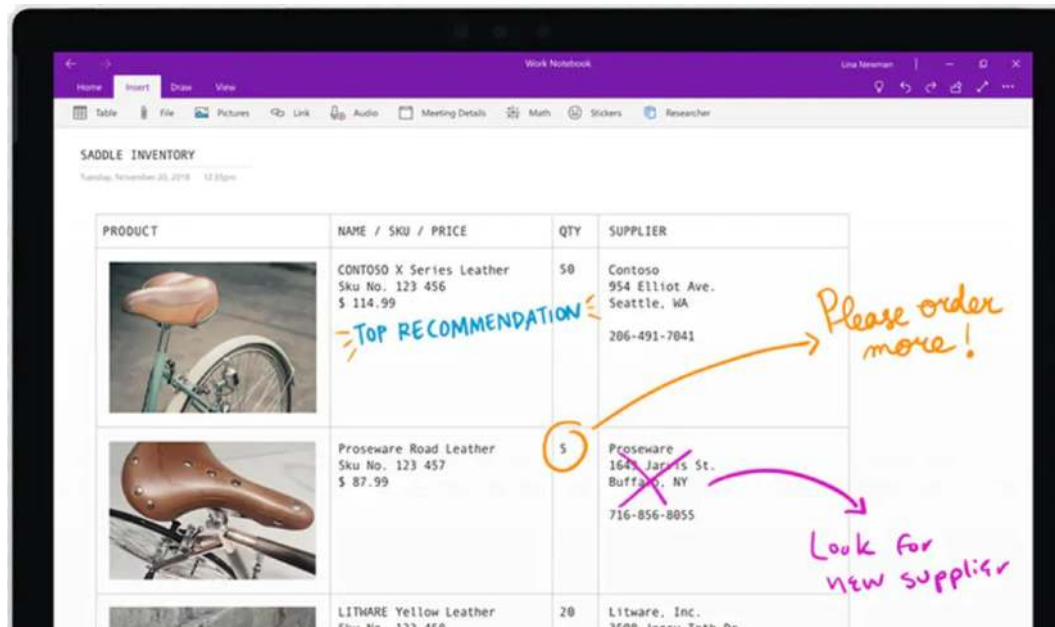
4.5 Note Taking

- **Note taking software** is an application that enables users to enter typed text, handwritten comments, drawings, sketches, photos, and links anywhere on a page and then save the page as part of a notebook.
- Some enable users to sync their notes to the cloud so that they can access the notes on any computer or mobile device. Many note taking applications also include a calendar feature.

4.5 Note Taking



Notes : Apple



Microsoft OneNote

4.6 Calendar and Contact Management

- **Calendar and contact management software** is an application that helps you organize your calendar, keep track of contacts, and share this information with other users, who can view it on their computers and mobile devices.
- This software provides a way for individuals and workgroups to organize, find, view, and share appointment and contact information easily.

Google Calendar

Apple Calendar

4.7 Software Suite

- A **software suite** is a collection of individual related applications available together as a unit. Productivity software suites typically include, at a minimum, word processing, presentation, spreadsheet, and email applications.
- Currently, Microsoft office is very popular example of software suite being used.

2011 - Microsoft Office 365 (Office 365)
2020 - Microsoft 365 (Home/Personal/Family)

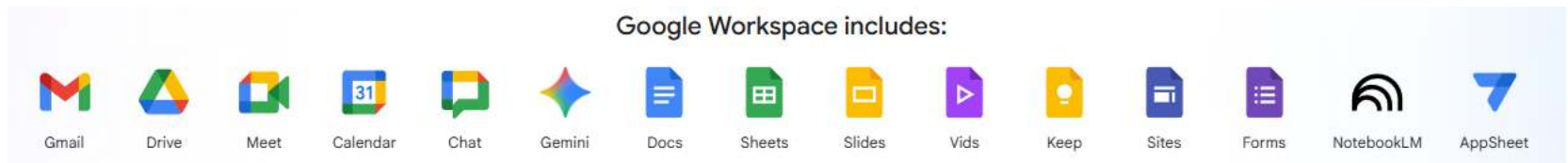


Microsoft Office 2024
Released



2006 - Google Apps -> G Suite
2020 - Google Workspace

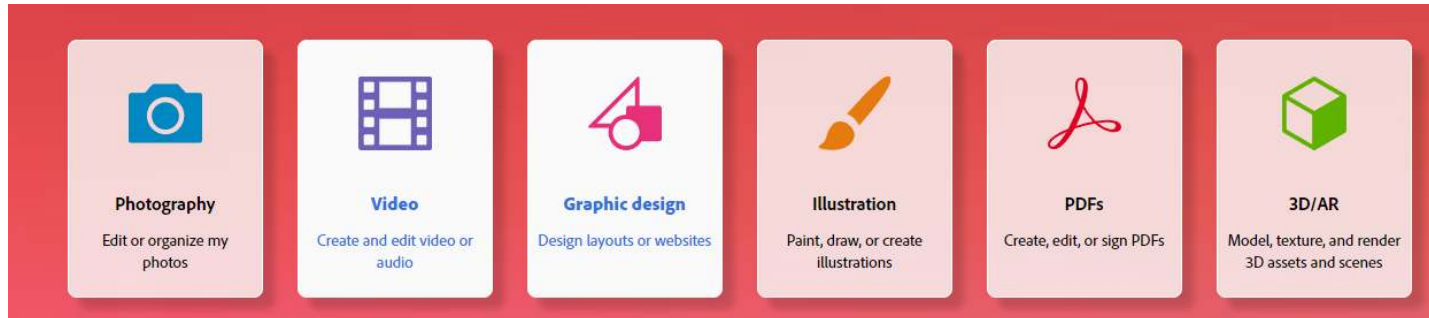
Google Workspace includes:



4.7 Software Suite



Creative Cloud



2013 - Adobe Creative Suite (CS) เปลี่ยนเป็น Adobe Creative Cloud (CC)



4.8 Project Management

- **Project management software** is an application that allows a user to plan, schedule, track, and analyze the events, resources, and costs of a project.
- Project management software helps users manage project variables, allowing them to complete a project on time and within budget.

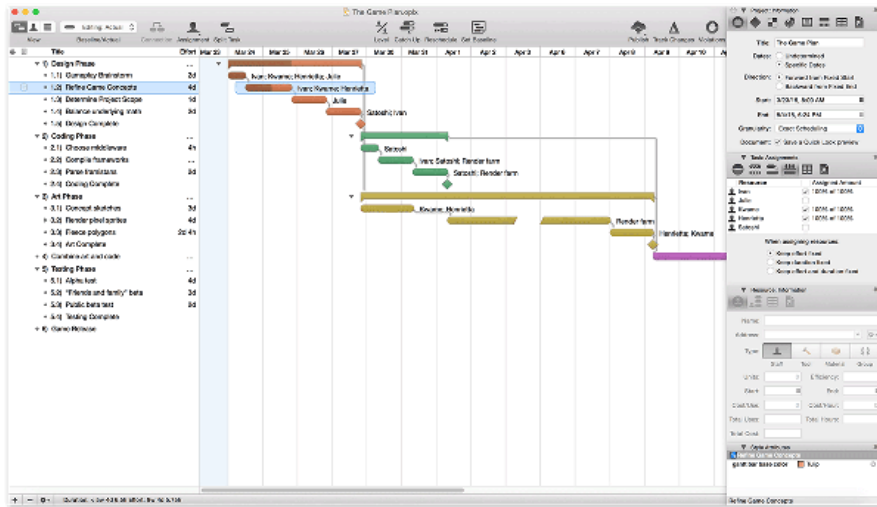
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4.8 Project Management

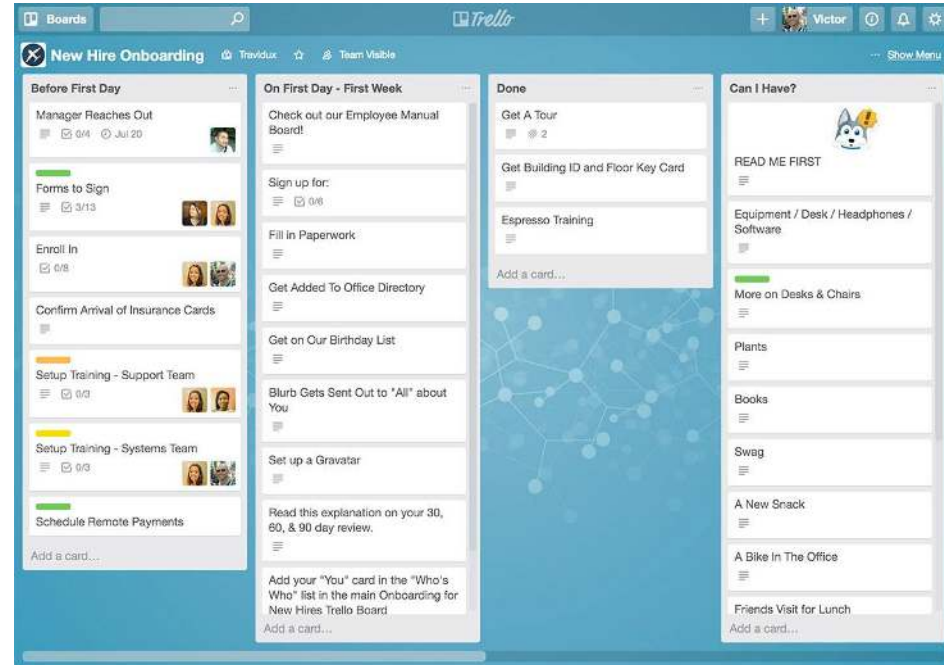
To manage a project using project management software, follow these steps:

- 1. Make sure you understand the project in its entirety**, as well as the steps you must take to bring the project to completion.
- 2. Determine the date** by which the project must be completed.
- 3. Verify you have the appropriate resources** (people and materials) **to complete the project**. If you do not have the necessary resources, you should obtain them.
- 4. Determine the order of the steps** that must be taken to bring the project to completion. Identify steps that must be taken before other steps, as well as steps that can be completed at the same time as other steps.
- 5. Verify the feasibility** of the plan.
- 6. During the project, it will be necessary to update the progress and possibly adjust dates.** Changes to the project and its dates should be communicated to the entire project team.

4.8 Project Management



Microsoft Project



4.8 Project Management Tools



Jira Software



4.9 Accounting

- **Accounting software** is an application that helps businesses of all sizes record and report their financial transactions. With accounting software, business users perform accounting activities related to the general ledger, accounts receivable, accounts payable, purchasing, invoicing, and payroll functions.

กระทำ

บัญชีแยกประเภท

ใบสั่งซื้อ

เงินเดือน

การซื้อ

- Most accounting software supports online credit checks, bill payment, direct deposit, and payroll services.

4.9 Accounting

Sample Rock Castle Construction - QuickBooks Accountant 2013 [US] - [Create Invoices]

File Edit View Lists Favorites Accountant Company Customers Vendors Employees Banking Reports Window Help

Home Calendar Snapshots Customers Vendors Employees Docs Reports Statement Writer App Center Invoice User Licenses

Main Formatting Send/Ship Reports

Find New Save Delete Create a Copy Memorize Mark As Pending Print Email Print Later Email Later Attach File Add Time/Costs Apply Credits Progress Receive Payments Create a Batch Refund/Credit Online Pay

CUSTOMER: JOB **Abercrombie, Kris** CLASS **Remodel** TEMPLATE **Rock Castl...**

Invoice

PAID 12/15/2017

DATE: 12/10/2017 INVOICE #: 1091

BILL TO: Kristy Abercrombie
5647 Cypress Hill Rd
Bayshore CA 94326

SHIP TO: Ship To 1
Kristy Abercrombie
5647 Cypress Hill Rd
Bayshore, CA 94326

TERMS: Net 30 DUE DATE: 01/09/2019

ITEM	DESCRIPTION	QUANTITY	U/M	RATE	AMOUNT	TAX
Removal	Removal labor	16		45.50	728.00	Non
Subs:Tile &C...	Install tile or counter			825.00	825.00	Non
Subs:Electrical	Electrical work			275.00	275.00	Non
Subs:Painting	Painting			154.00	154.00	Non
Subs:Plumbi...	Plumbing			550.00	550.00	Non

ONLINE PAY: Off TAX: San Tomas (7.75%) 0.00

TOTAL: 4,522.00

CUSTOMER MESSAGE: PAYMENTS APPLIED: 4,522.00

BALANCE DUE: 0.00

MEMO: CUSTOMER TAX CODE: Tax Save & Close Save & New Revert

Abercrombie, Kristy:Remodel B...

Customer Transaction

SUMMARY

Phone: 415-555-6579
Email: kristy@samplename.com
Open balance: 0.00
Active estimates: 1
Sales Orders to be invoiced: 0
Unbilled time and expenses: 1,910.00

RECENT TRANSACTION

12/15/17 Payment	7,633.28
12/12/17 Estimate	7,676.13
12/10/17 Invoice - Paid	4,522.00
12/01/17 Credit Memo	711.15
11/25/17 Invoice - Paid	3,111.28

NOTES

4.10 Personal Finance

- **Personal finance** software is a simplified accounting application that helps home users and small/home office users balance their checkbooks, pay bills, track personal income and expenses, verify account balances, transfer funds, track investments, and evaluate financial plans.

ประเมิน

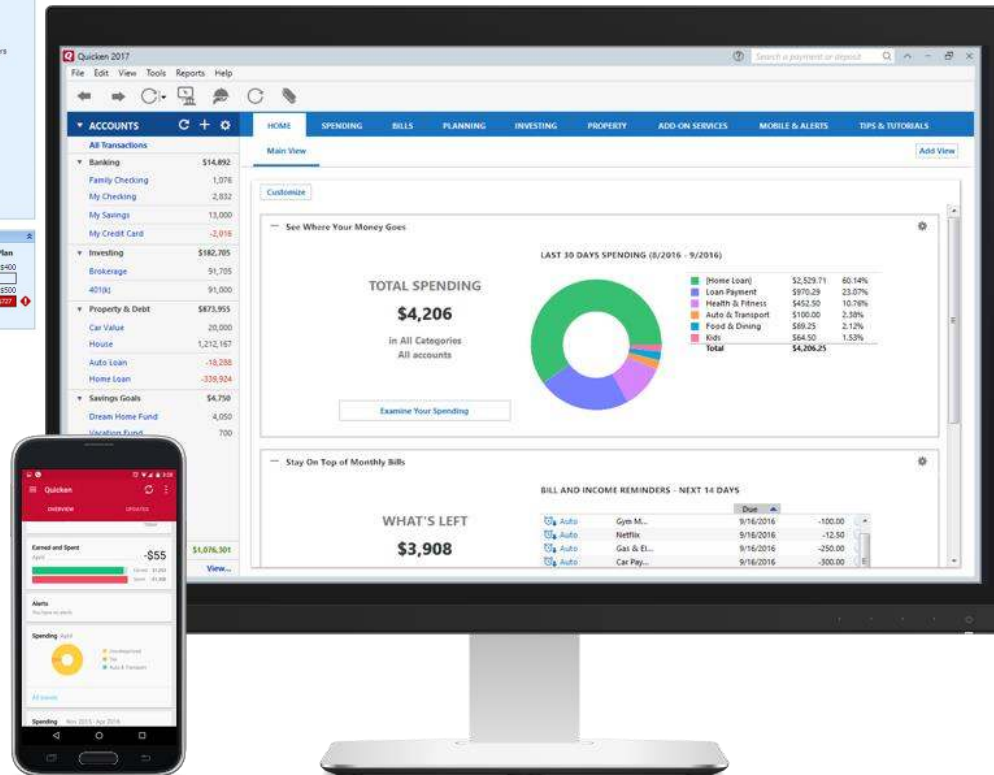
- Personal finance software helps determine where, and for what purpose, you are spending money so that you can manage your finances.

4.10 Personal Finance

Quicken



Microsoft Money



4.11 Legal

- **Legal software** is an application that assists in the preparation of legal documents and provides legal information to individuals, families, and small businesses. แต่ละบุคคล
- Legal software provides standard contracts and documents associated with buying, selling, and renting property; estate planning; marriage and divorce; and preparing a will or living trust. ทรัพย์สิน ทรัพย์สินที่ดิน หนี้กรรม ผู้จัดการทรัพย์สิน

4.11 Legal

Take your practice to the next level with the most
intuitive trust accounting software.



MyAttorney HOME & BUSINESS

Get Started Choose Docs Create Docs Manage Docs Advice Library Finances

Interview: Bill of Sale - Motor Vehicle [Show Preview](#) [Show Help](#) [Close Interview](#)

What motor vehicle is being sold?

Year

Make

Model

VIN

For example, 1990, Toyota, Camry, 2B9A5436BH5868.

[First](#) [Back](#) [Next](#) [Bookmark](#)

Bill of Sale - Motor Vehicle

The Bill of Sale - Motor Vehicle is a document under which one or more sellers transfer title rights in a motor vehicle to one or more buyers. This document only represents the transfer of the RIGHT to ownership; the motor vehicle's certificate of title represents ACTUAL ownership. Thus, this document and the certificate of title are needed to complete the transfer of the motor vehicle.

The Bill of Sale - Motor Vehicle should be signed by the seller(s), and becomes effective as of the date provided in the text of the document. The buyer(s) should sign the Bill of Sale - Motor Vehicle immediately below the odometer statement in the document. The buyer(s) should also

BILL OF SALE - MOTOR VEHICLE

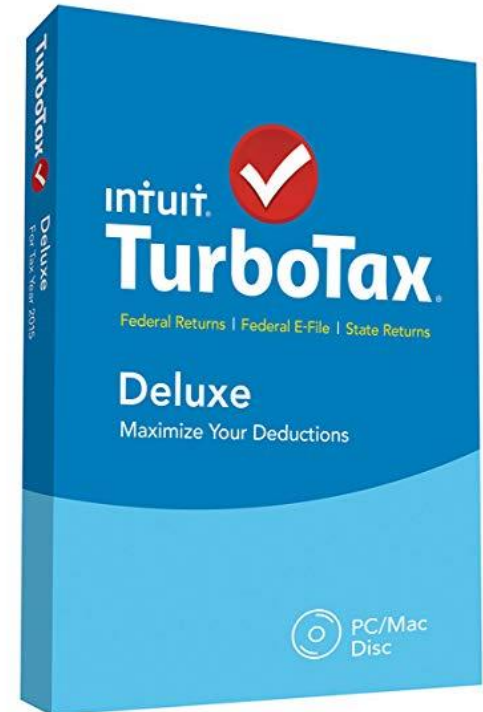
For the consideration of \$5,250.00, Sara Albertson (the "Seller"), of 2887 Fourth Place, Harrington, Indiana 46322, does hereby sell, assign and transfer to Jesse Martinez (the "Buyer"), of 7761 Cedar Road, Riverton, Indiana 46323, the following described motor vehicle (the

4.12 Tax Preparation

- **Tax preparation** software is an application that can guide individuals, ^{แต่ละบุคคล} families, or small businesses through the process of filing federal and state taxes. ^{สหพันธรัฐ,}
- These programs forecast tax liability and offer money-saving tax tips, designed to lower your tax bill. ^{หนี้สิน}
- After you answer a series of questions and complete basic forms, the software creates and analyzes your tax forms to search for missed potential errors and deduction opportunities.

4.12 Tax Preparation

The screenshot displays the TaxACT Deluxe Edition software interface. The main window is titled "Federal Form 1040" and shows the "U.S. Individual Income Tax Return" form. The left sidebar lists various forms and schedules, including Form 1040, Form 1040A, Form 1040EZ, Form 1040-SS, Form 1040V, Form 1040-ES, Form 1040X, Schedule A, Schedule B, Schedule C, Schedule D, Schedule E, Schedule F, Schedule H, Schedule J, Schedule R, Schedule SE, Form 1045, Form 1045-S, Form 1116, Form 1310, Form 2106, Form 2120, Form 2210, and Form 2210-F. The main form area shows the "Filing Status" section with options for Single, Married filing joint return, Married filing separate return, Head of household, and Qualifying widow(er). The "Student Status" section includes checkboxes for "Taxpayer was a student" and "Spouse was a student". The "Disability Status" section includes checkboxes for "Taxpayer was disabled" and "Spouse was disabled". The "Exemptions" section is partially visible at the bottom.

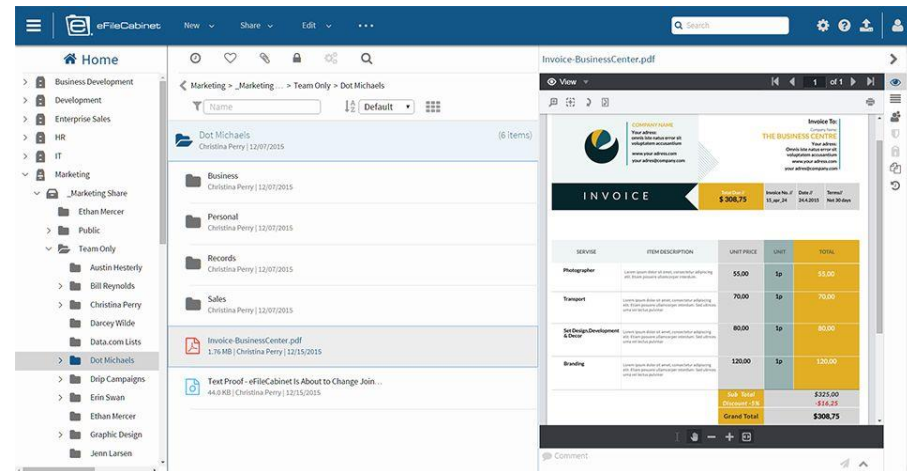
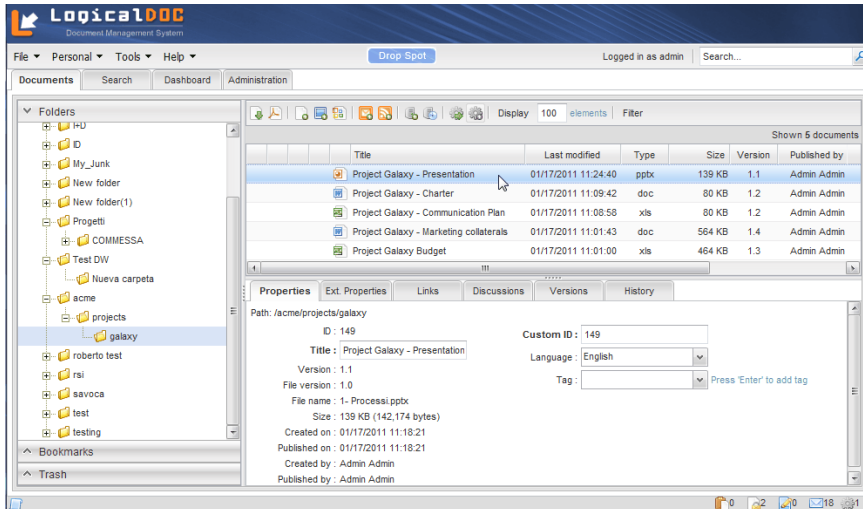
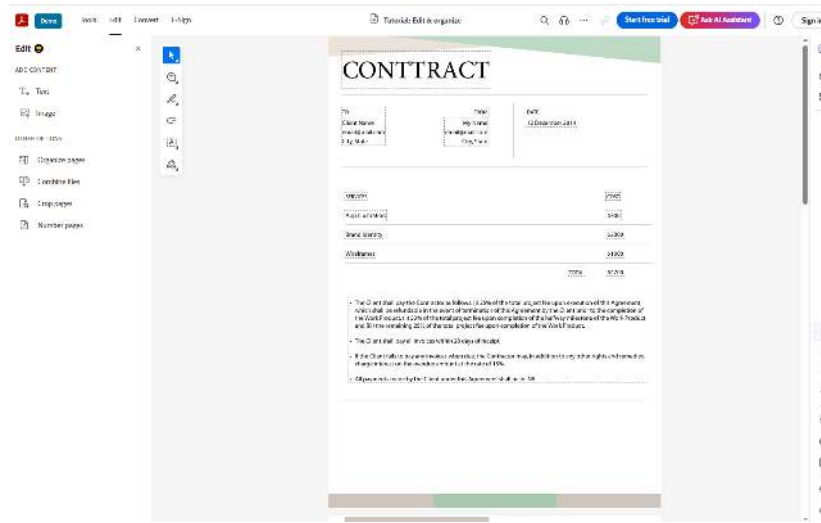


TaxAct

4.13 Document Management

- **Document management** software is an application that provides a means for sharing, distributing, and searching through documents by converting them into a format that can be viewed by any user.
- The converted document, which mirrors the original document's appearance, can be viewed and printed without the software that created the original document.
- Some document management software allows users to edit - content and add comments to the converted document.
- A popular electronic image file format that document management software uses to save converted documents is **PDF (Portable Document Format)**, developed by Adobe Systems.

4.13 Document Management



4.14 Enterprise Computing

- A large organization, commonly referred to as an enterprise, requires special computing solutions because of its size and geographic distribution.
- A typical enterprise consists of a wide variety of departments, centers, and divisions — collectively known as functional units.

Human resources
software

Accounting
software

Engineering or
product
development
software

Manufacturing
software

Marketing software

Sales software

Distribution
software

Customer service
software

Software for
information
technology staff

5. Graphics and Media Applications

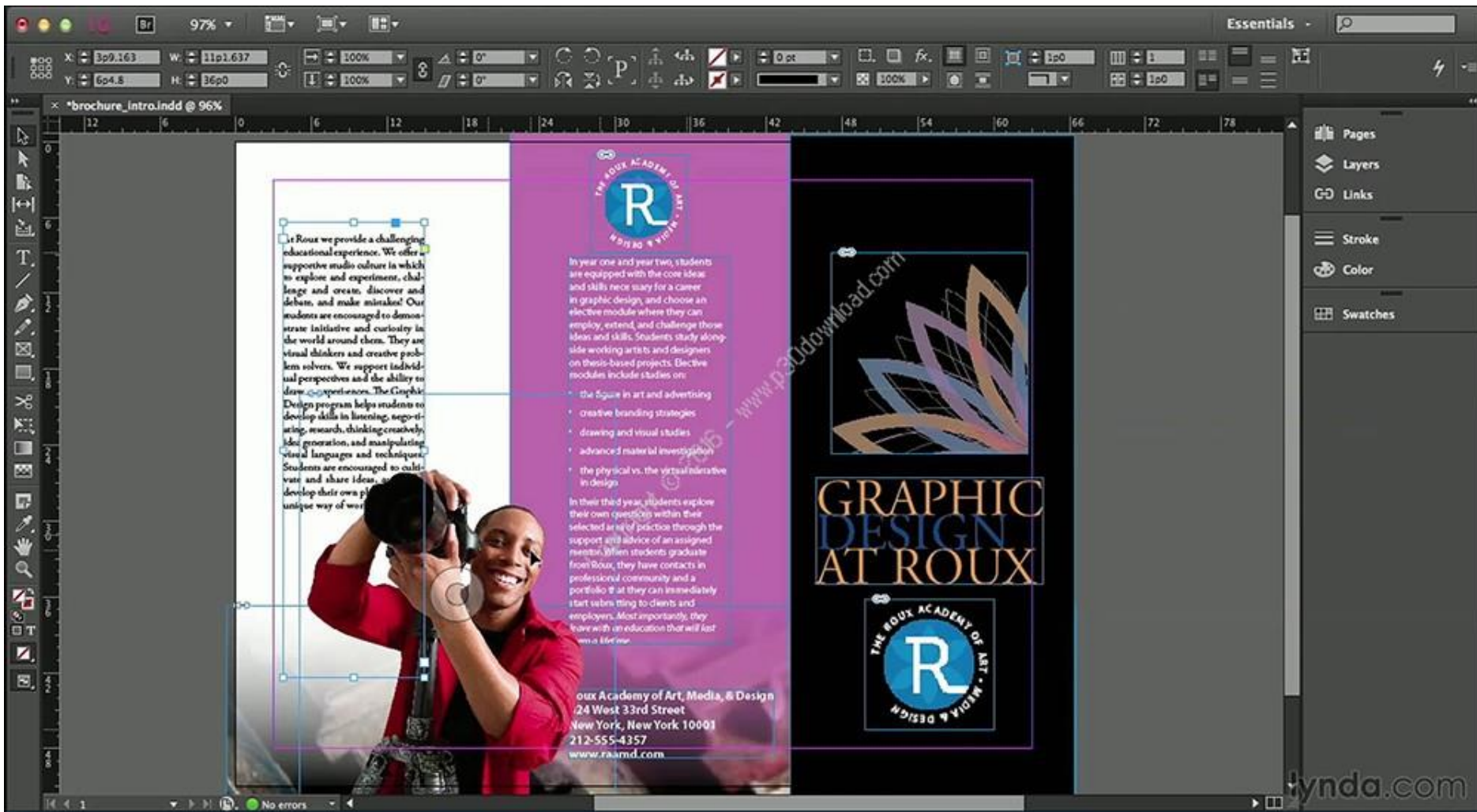
5.1 Computer-aided design (CAD) software is a type of application that assists professionals and designers in creating engineering, architectural, and scientific designs and models. Three-dimensional CAD programs allow designers to rotate designs of 3-D objects to view them from any angle.



5.2 Desktop Publishing

- **Desktop publishing software** (DTP software) is an งานที่ซับซ้อน application that enables designers to create sophisticated publications that contain text, graphics, and many colors.
- Professional DTP software is ideal for the production of high-quality color projects such as textbooks, corporate newsletters, marketing literature, product catalogs, and annual reports.
- Designers and graphic artists can print finished publications on a color printer, take them to a professional printer, or post them on the web in a format that can be viewed by those without DTP software.

5.2 Desktop Publishing

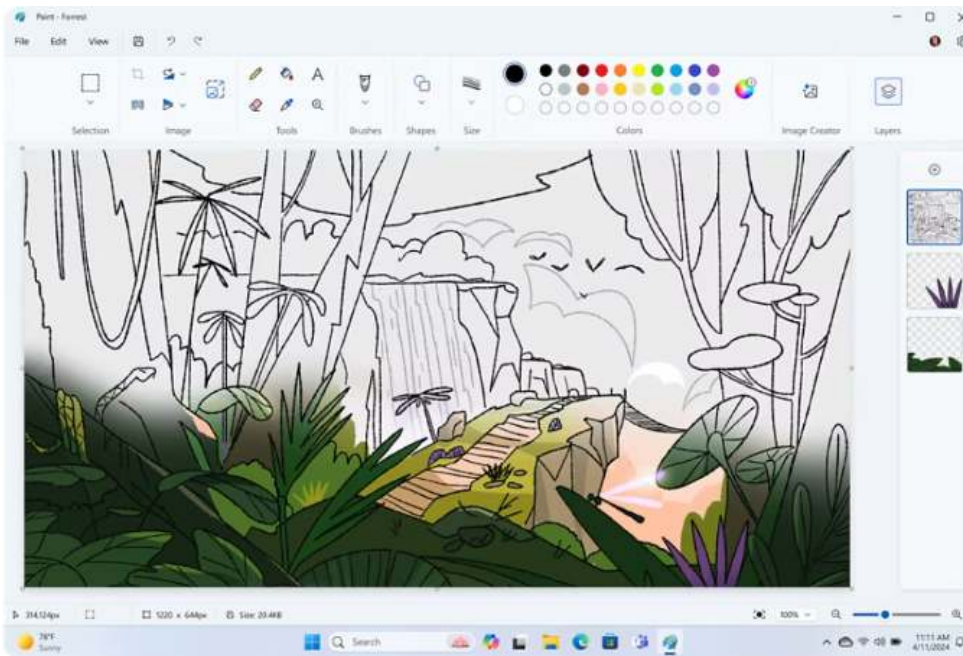


5.3 Paint/Image Editing

- **Paint software**, also called **illustration software**, is an application that allows users to draw pictures, shapes, and other graphics with various on-screen tools, such as a pen, brush, eyedropper, and paint bucket.
- **Image editing** software is an application that provides the capabilities of paint software and also includes the capability to enhance and modify existing photos and images.
- Modifications can include adjusting or enhancing image colors, adding special effects such as shadows and glows, creating animations, and image stitching (combining multiple images into a larger image).

5.3 Paint/Image Editing

Microsoft Paint



Source: <https://www.microsoft.com/en-us/windows/paint#backgroundremoval>

Sketchup (by Trimble)



Source: <https://sketchup.trimble.com/en/whats-new>

5.4 Photo Editing and Photo Management

- **Photo editing software** is a type of image editing software that allows users to edit and customize digital photos.
- With photo editing software, users can retouch photos, crop images, remove red-eye, erase blemishes, restore aged photos, add special effects, enhance image quality, change image shapes, color-correct images, straighten images, remove or rearrange objects in a photo, add layers, and more.
- Many applications also provide a means for creating digital photo albums.

5.4 Photo Editing and Photo Management

ACTION	PURPOSE
<i>Crop</i>	Removes unwanted areas of a photo
Remove <i>red-eye</i>	Removes the appearance of red eyes caused by the camera flash
<i>Resize</i>	Changes the physical dimensions of the photo
<i>Compress</i>	Decreases the photo's file size
Adjust <i>sharpness</i>	Increases or decreases crispness of objects in the photo
Adjust <i>brightness</i>	Adjusts lightness or darkness in the photo
Adjust <i>contrast</i>	Adjusts the difference in appearance between light and dark areas of the photo

5.4 Photo Editing and Photo Management



Before



After

Adjust sharpness



Before



After Adjust contrast to 89 %

Adjust contrast

5.4 Photo Editing and Photo Management

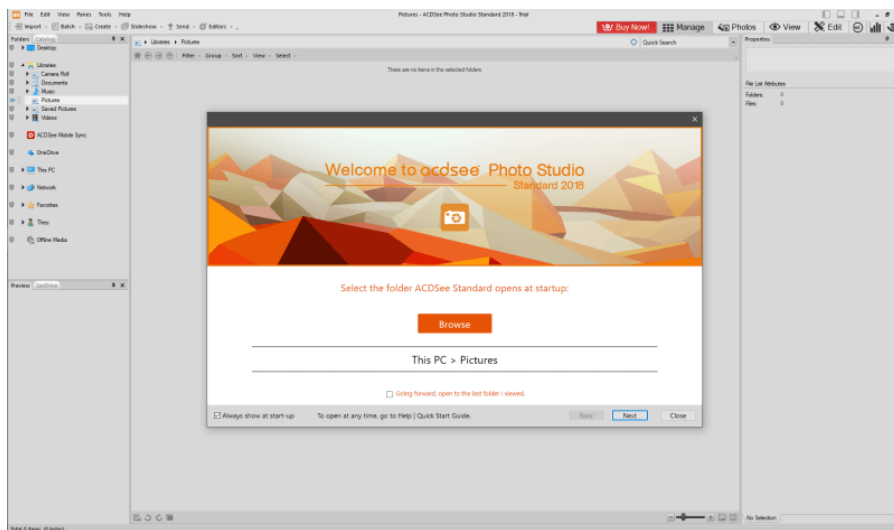


5.4 Photo Editing and Photo Management

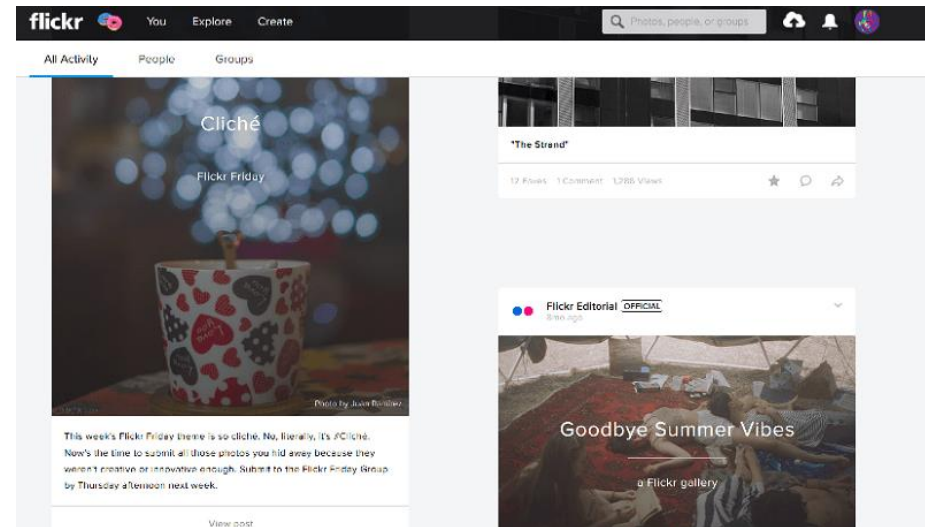
- With **photo management software**, you can view, organize, sort, catalog, print, and share digital photos. Some photo editing software includes photo management functionality.
- Many online photo storage services enable you to create scrapbooks — selecting photos, adding captions, selecting backgrounds, and more.

5.4 Photo Editing and Photo Management

ACDSee Photo Studio Standard 2018



Flickr



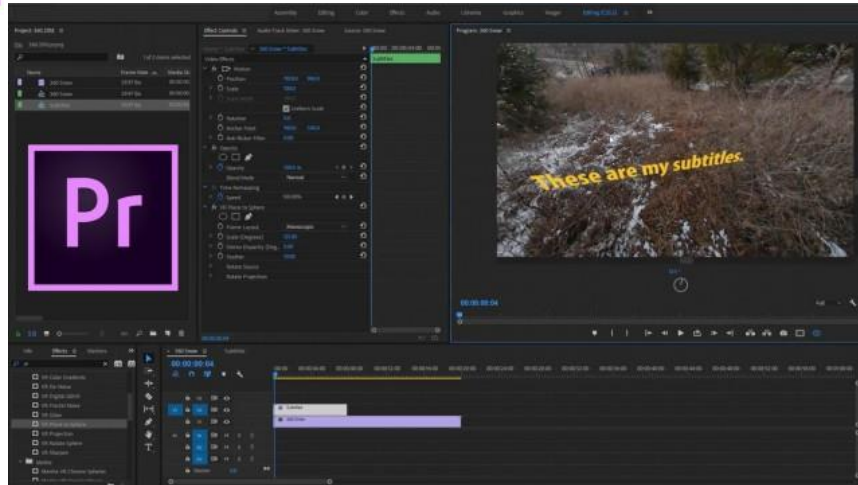
5.5 Video and Audio Editing

- **Video editing** software is an application that allows professionals to modify a segment of a video, called a clip. For example, users can reduce the length of a video clip, reorder a series of clips, or add special effects such as words that move across the screen. Video editing software typically includes audio editing capabilities.
- **Audio editing** software is an application that enables users to modify audio clips, produce studio-quality soundtracks, and add audio to video clips.

5.5 Video and Audio Editing



Adobe Premiere Pro



Vegas Pro



Final Cut Pro

5.6 Multimedia and Website Authoring

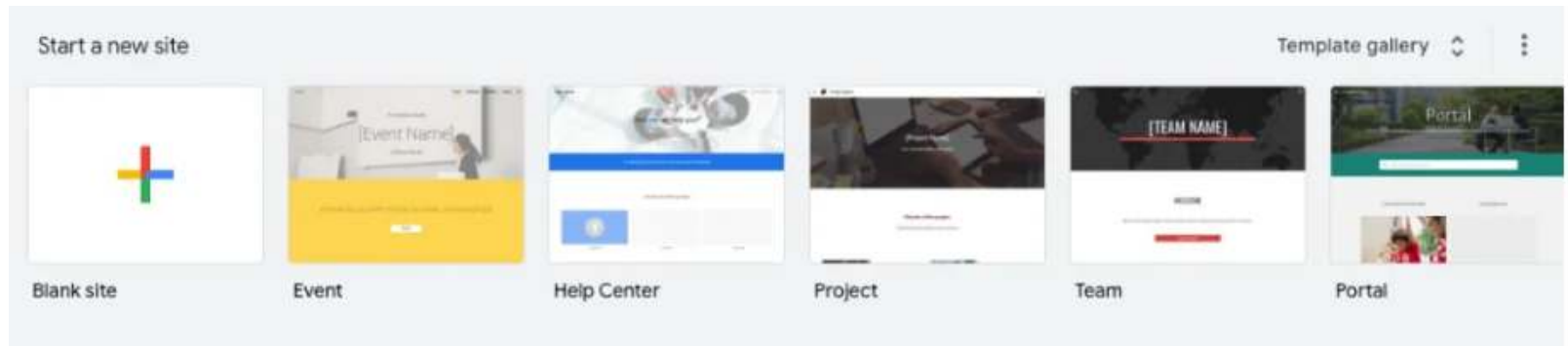
ซอฟต์แวร์สำหรับสร้างสื่อมัลติมีเดีย

- **Multimedia authoring software** allows users to combine text, graphics, audio, video, and animation in an interactive application. With this software, users control the placement of text and images and the duration of sounds, video, and animation.

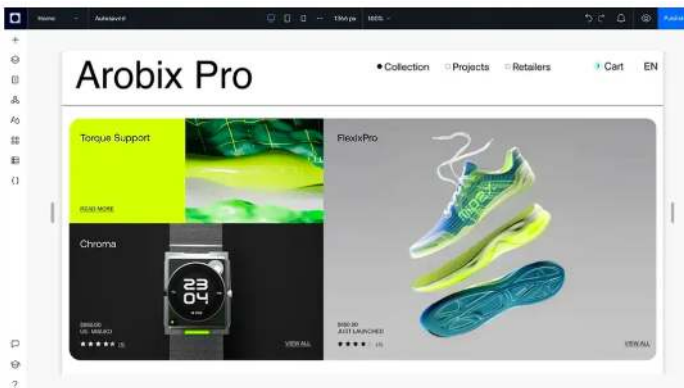
ซอฟต์แวร์สำหรับสร้างเว็บไซต์

- **Website authoring software** helps users of all skill levels create related webpages that include graphics, video, audio, animation, special effects with interactive content, and blog posts. In addition, many website authoring programs allow users to organize, manage, and maintain websites.

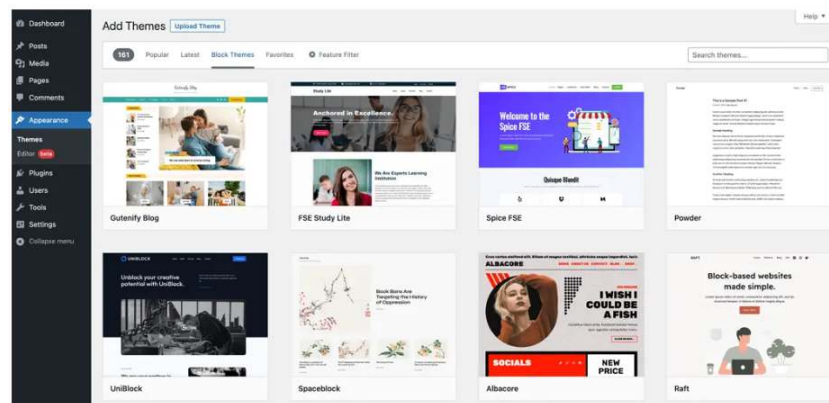
Google Site



Wix



Wordpress

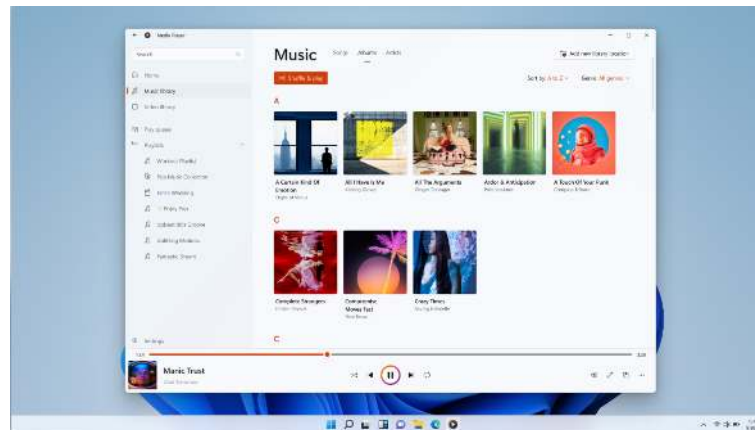


Multimedia and Website Authoring

Sourec: <https://www.neumerlingroup.com/story/website-development-tools>

5.7 Media Player

- A **media player** is a program that allows you to view images and animations, listen to audio, and watch video files on your computer or mobile device.
- Media Player is "designed to make listening to and watching your multimedia content more enjoyable. At the heart of Media Player is a full-featured music library that allows you to quickly browse and play music, as well as create and manage playlists. All your content in the music and video folders on your PC will appear automatically in your library."



6. Personal Interest Applications

Most of the programs in this category are relatively inexpensive; many are free. Some applications focus on a single service, while others provide several services in a single application.

Lifestyle
applications

Medical
applications

Entertainment
applications

Convenience
applications

Education
applications

6. Personal Interest Applications

รูปแบบการดำเนินชีวิต

- **Lifestyle applications:** Access the latest news or sports scores, check the weather forecast, compose music, research genealogy, find recipes, or locate nearby restaurants, gas stations, or points of interest.

อาการของโรค

- **Medical applications:** Research symptoms, establish a fitness or health program, track exercise activity, refill prescriptions, count calories, or monitor sleep patterns.

เติมอีก

ใบสั่งยา

- **Entertainment applications:** Listen to music or the radio, view photos, watch videos or shows, read a book or other publication, organize and track fantasy sports teams, and play games individually or with others.

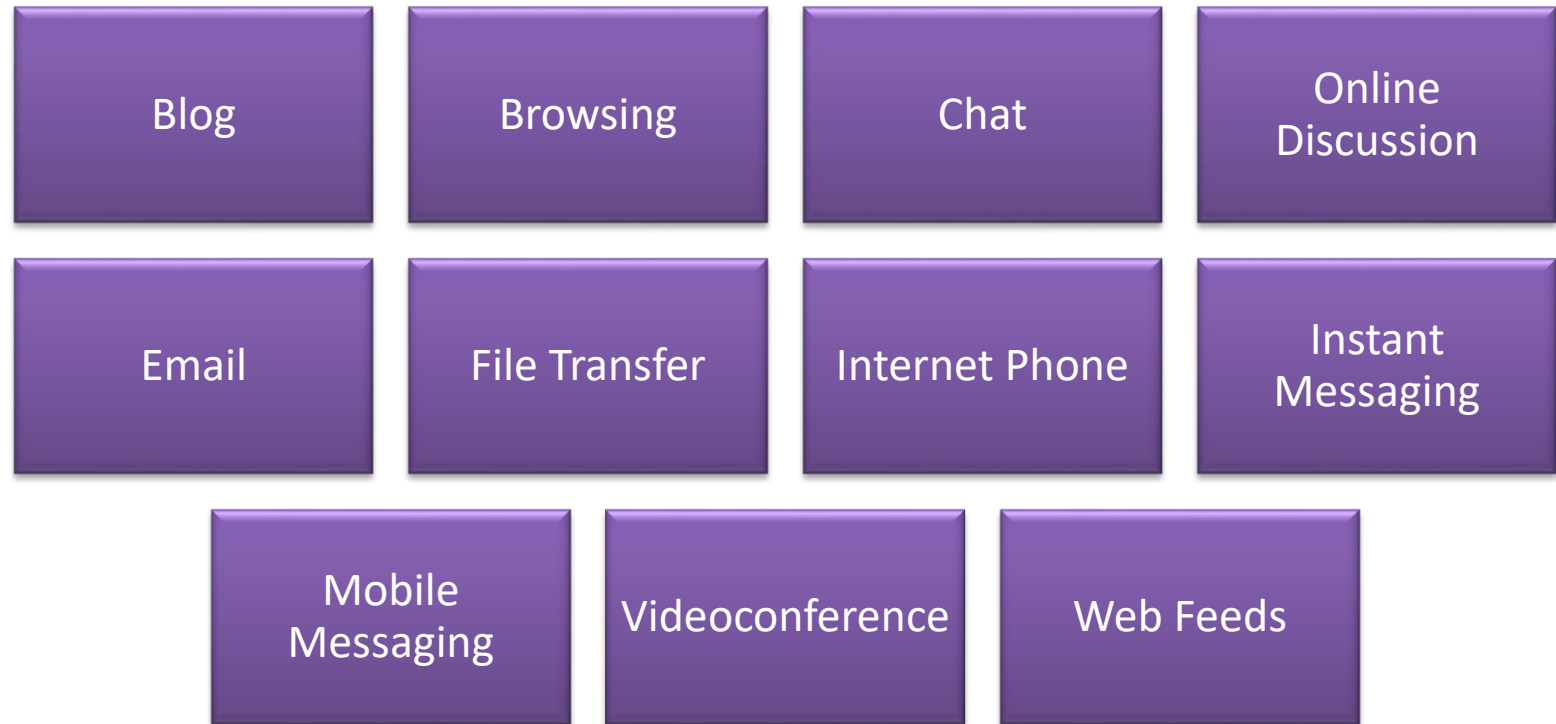
ความสะดวก

- **Convenience applications:** Obtain driving directions or your current location, remotely start your vehicle or unlock/lock the doors, set an alarm or timer, check the time, calculate a tip, use your phone as a flashlight, or use a personal assistant that acts on your voice commands.

- **Education applications:** Access how-to guides, learn or fine-tune a particular skill, follow a tutorial, run a simulation, assist children with reading and other elementary skills, or support academics.

7. Communications Applications

One of the main reasons people use computers is to communicate and share information with others.



7. Communications Applications

- ❑ **Blog** Time-stamped articles, or posts, in diary or journal format, usually listed in reverse chronological order.
ตามลำดับเวลา
- ❑ **Browsing** Allows users to access and view webpages on the Internet
- ❑ **Chat** Real-time, online typed conversation with one or more users
- ❑ **Online Discussion** Online areas where users have written discussions
- ❑ **Email** Messages and files sent via a network, such as the Internet

7. Communications Applications

- ❑ **File Transfer** Method of uploading files to and downloading files from servers on the Internet
- ❑ **Internet Phone** Allows users to speak to other users via an Internet connection
- ❑ **Internet Messaging** Real-time exchange of messages, files, images, audio, and/or video with another online user
- ❑ **Mobile Messaging** Short text, picture, or video messages sent and received, mainly on mobile devices
- ❑ **Videoconference** Meeting between geographically separated people who use a network to transmit video/audio
- ❑ **Web Feeds** Keeps track of changes made to blogs by checking feeds

8. Security Tools

Virus

ความตั้งใจ The programmer of a virus, known as a **virus author**, intentionally writes a virus program. Writing a virus program usually requires significant programming skills. The virus author ensures the virus can replicate itself, conceal itself, monitor for certain events, and then deliver its payload.

The first step in the **infection phase** is activation of the virus. The most common way viruses spread is by users running infected programs or apps.

1. First, a virus replicates by attaching itself to program files.
2. Viruses ^{ปกปิด} conceal themselves to avoid detection.
3. Finally, viruses watch for a certain condition or event and activate when that condition or event occurs. *The event might be starting the computer or device, or reaching a date on the system clock.*

8. Security Tools

Virus

During the **delivery phase**, the virus unleashes its payload, which might be a harmless prank that displays a meaningless message — or it might be **destructive, corrupting or deleting data and files.**

ปล่อย

การล้อเล่น

ซึ่งเป็นการทำลาย

ทำให้เสื่อม

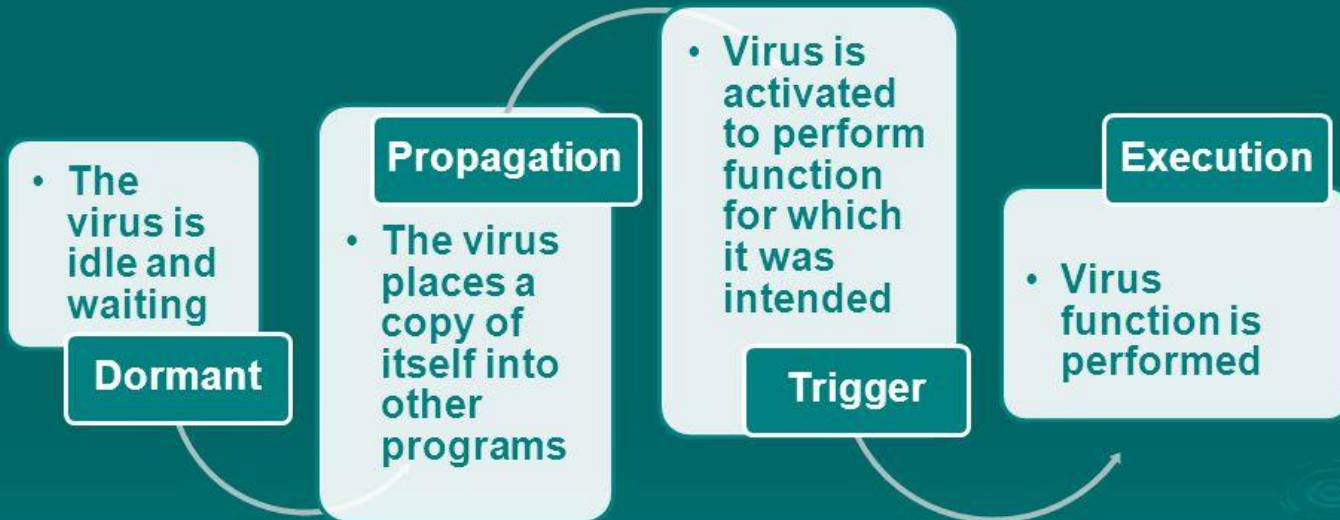
The most dangerous viruses do not have an obvious payload. Instead, they quietly modify files.

อย่างเห็นชัด

One way antivirus software detects computer viruses is by monitoring files for unknown changes.

8. Security Tools

Virus Lifecycle

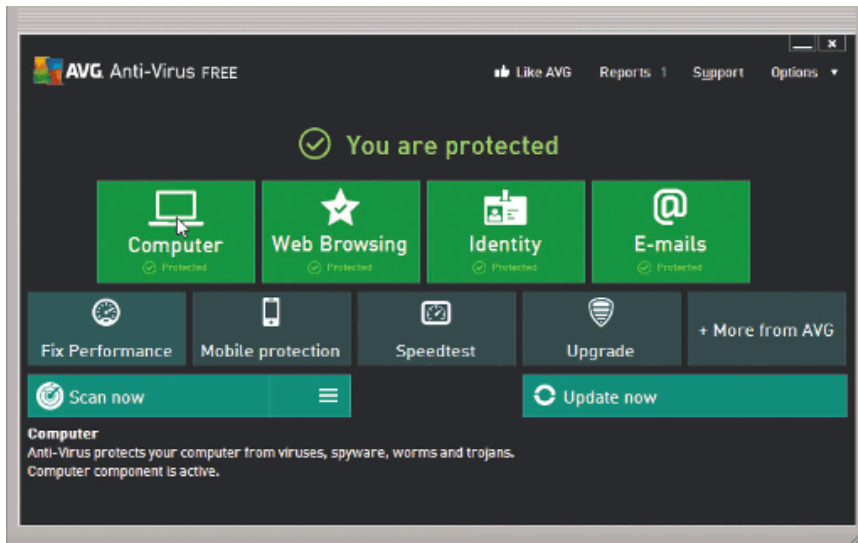


8. Security Tools

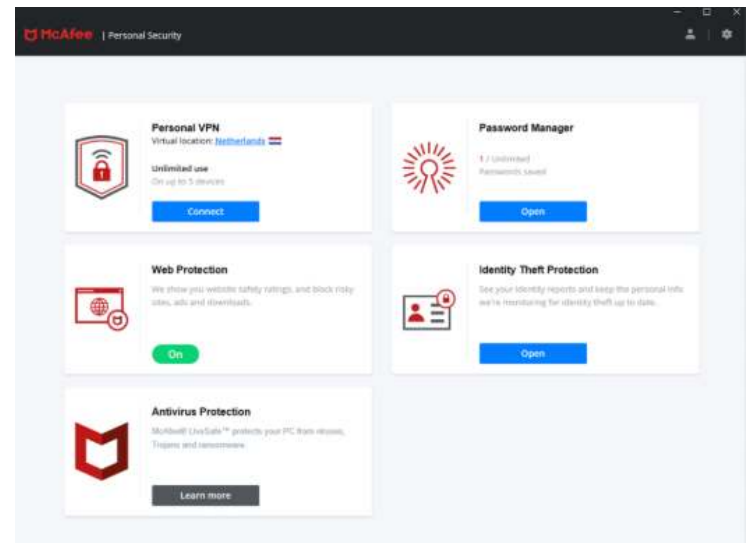
- In addition to viruses, other malware includes worms, trojan horse programs, and rootkits.
- **A worm** resides in active memory and replicates itself over a network to infect computers and devices, using up system resources and possibly shutting down the system.
- **A trojan horse** is a destructive program disguised as a real program, such as a screen saver. When a user runs a seemingly innocent program, a trojan horse hiding inside can capture information, such as user names and passwords, from your computer or enable someone to control your computer remotely. Unlike viruses, trojan horses do not replicate themselves.
- **A rootkit** is a program that easily can hide and allow someone to take full control of your computer from a remote location, often for nefarious purposes. For example, a rootkit can hide in a folder on your computer. The folder appears empty because the rootkit has instructed your computer not to display the contents of the folder. Rootkits can be very dangerous and often require special software to detect and remove.

8.1 Antivirus program

- An **antivirus program** protects a computer against viruses by identifying and removing any computer viruses found in memory, on storage media, or on incoming files



AVG Antivirus

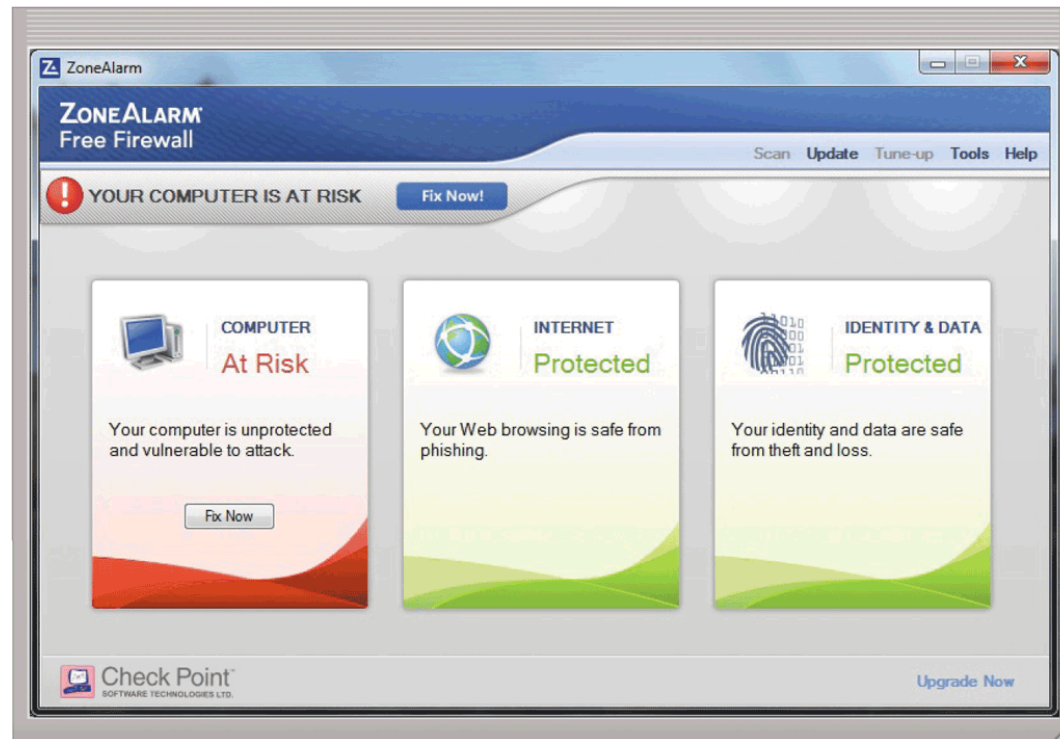


McAfee Antivirus

8.2 Personal firewall

- A **personal firewall** is a security tool that detects and protects a personal computer and its data from unauthorized intrusions

ការປ້ອງກັນ,



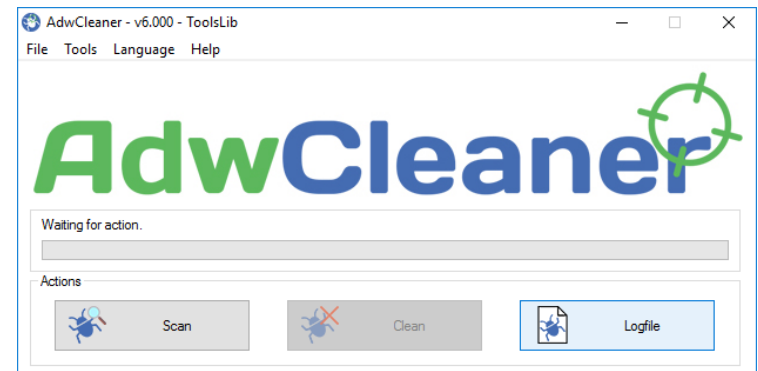
8.3 Spyware & Adware

- **Spyware** is a type of program placed on a computer or mobile device without the user's knowledge that secretly collects information about the user and then communicates the information it collects to some outside source while the user is online
- **Adware** is a type of program that displays an online advertisement in a banner or pop-up or pop-under window on webpages, email messages, or other Internet services

8.3 Spyware & Adware

A **spyware remover** is a type of program that detects and deletes spyware and similar programs

An **adware remover** is a program that detects and deletes adware



8.4 Internet filter

- Filters are programs that remove or block certain items from being displayed.

**Anti-spam
programs**

**Web filtering
software**

**Phishing
filters**

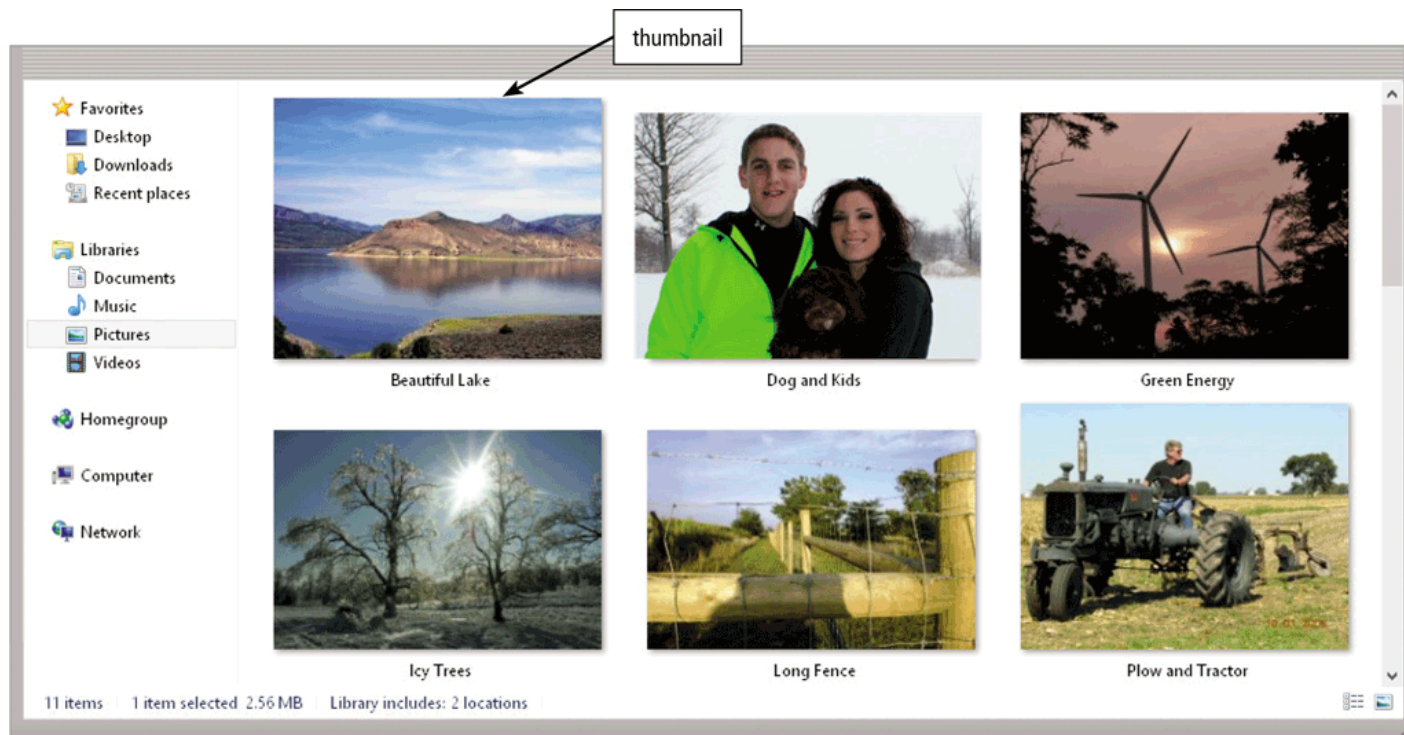
**Pop-up and
pop-under
blockers**

8.4 Internet filter

- **Anti-Spam Programs** Spam is an unsolicited email message or posting sent to many recipients or forums at once. ไม่ได้เรียกร้อง
- **Web Filters** Web filtering software is a program that restricts access to certain material on the web.
- **Phishing Filters** Phishing is a scam in which a perpetrator sends an official looking email message that attempts to obtain your personal and/or financial information. เล่ห์อุบาย ผู้ทำชั่ว, ผู้กระทำผิด
- **Pop-Up and Pop-Under Blockers** A pop-up ad is an Internet advertisement that suddenly appears in a new window on top of a webpage.

9. File, Disk, and System Management Tools

- A **file manager** is a tool that performs functions related to file management



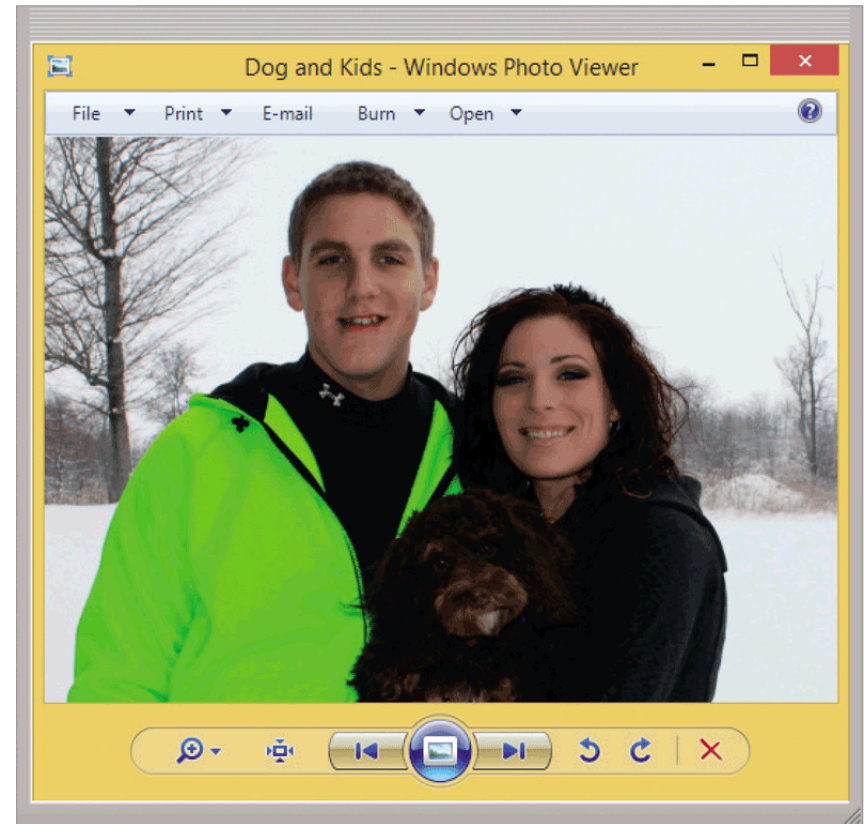
9. File, Disk, and System Management Tools

- A **search tool** is a program, usually included with an operating system, that attempts to locate a file, contact, calendar event, app, or any other item stored on your computer or mobile device based on criteria you specify



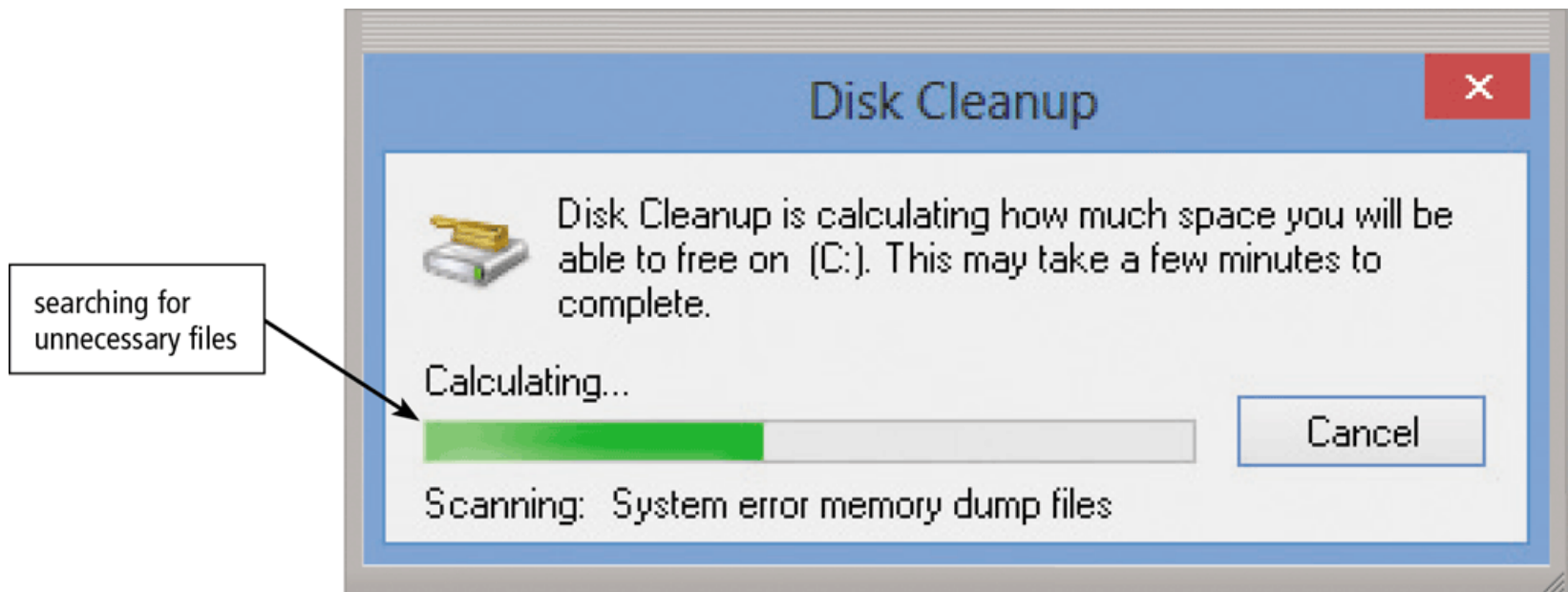
9. File, Disk, and System Management Tools

- An **image viewer** is a tool that allows users to display, copy, and print the contents of a graphics file, such as a photo
- An **uninstaller** is a tool that removes a program, as well as any associated entries in the system files



9. File, Disk, and System Management Tools

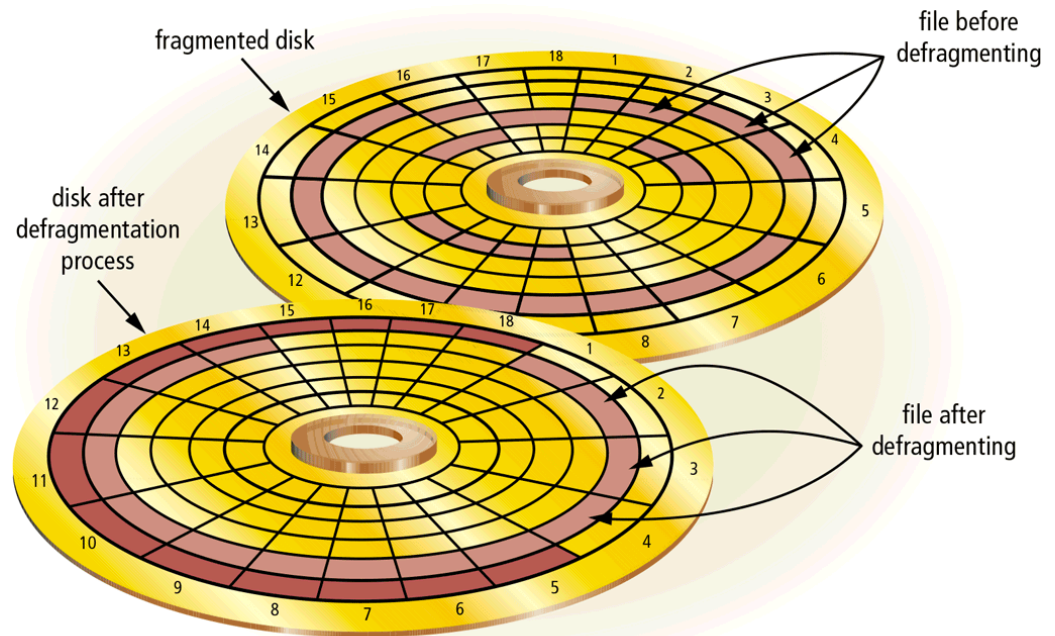
- A **disk cleanup** tool searches for and removes unnecessary files



9. File, Disk, and System Management Tools

เครื่องมือช่วยการจัดเรียงข้อมูลใน hard drive ใหม่ เพื่อให้ OS สามารถเข้าถึงข้อมูลได้ไวขึ้น

- A **disk defragmenter** is a tool that reorganizes the files and unused space on a computer's hard disk so that the operating system accesses data more quickly and programs run faster
 - Defragmenting



9. File, Disk, and System Management Tools

A **screen saver** is a tool that causes a display device's screen to show a moving image or blank screen if no keyboard or mouse activity occurs for a specified time

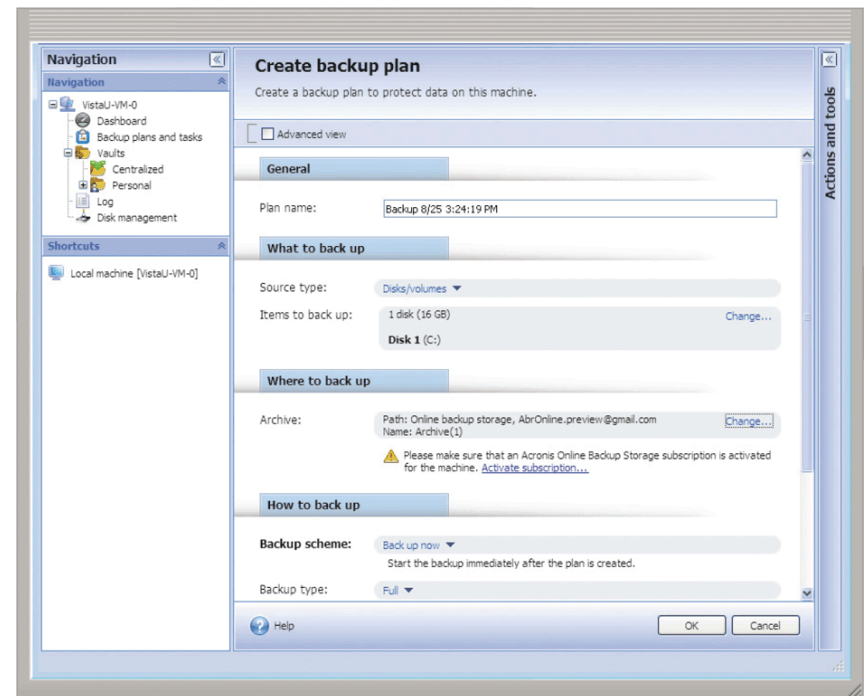
บีบอัดให้มีขนาดเล็กลง

A **file compression tool** shrinks the size of a file(s)

A **PC maintenance tool** is a program that identifies and fixes operating system problems, detects and repairs drive problems, and includes the capability of improving a computer's performance

9. File, Disk, and System Management Tools

- A **backup tool** allows users to copy, or back up, selected files or the contents of an entire storage medium to another storage location
- A **restore tool** reverses the process and returns backed up files to their original form



Any Question ?

